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# An Approach towards an Efficient Encryption-Decryption of Grayscale and Color Images

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**Abstract**

Protecting images from unauthorized access and use is an important task in various areas. The problem of unauthorized use of images is solved at different levels: from the provisions on copyright, to methods of cryptography and steganography, digital watermarking etc. The paper considers a new, authors-developed approach to encryption-decryption of grayscale and color images, which is based on the RSA algorithm. We describe the algorithmic implementation of encrypting and decrypting procedures by one row of image matrix with and without use of additional noise-adding. We developed the procedures of encrypting-decrypting of images by one row of image matrix by combining elements of the RSA algorithm and bitwise binary operations. The effectiveness of the proposed approach is confirmed by conducting a series of experiments with grayscale and color images. We also study the influence of additional noise-adding functions applied to the source image as well as the various values of simple numbers of the RSA system on the results of the method. It was found that these parameters do not result in appearance of contours in the encrypted image, and the resulting image differs from the source one by a small change in the brightness level only. In addition to the high cryptographic strength of the RSA algorithm, the overall method's cryptostrength is provided by the additional use of bitwise binary operations.

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*Keywords:* Encrypting; decrypting; image; edge; cryptographic stability; affine transformation;

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## 1. Introduction

The main basis of image protection is the assumption that the image is a stochastic signal [1]. However, in addition to typical informativity (data informativeness), image is also a specific signal that possesses visual informativeness [2]. Images as stochastic signals are among the most commonly used types of information [3]. Accordingly, the urgent task is to protect such images from unauthorized access and use [4-8]. This leads to the use of known classic encryption methods in the case of image encryption [2, 9].

We propose a new image encryption-decryption algorithm, developed on the basis of the RSA algorithm elements [9], which is effective in case of images with clearly defined contours. RSA algorithm elements are used as coefficients of some affine transformations. Proposed modification has a higher cryptographic stability than RSA algorithm [2, 9]. It is also describing the usage of elements of the RSA algorithm in affine transformations for image encrypting-decrypting. In relation to image encryption, the actual task [4] is the realization of such application of the RSA algorithm, that:

- will not reduce the cryptographic stability of the RSA algorithm;
- will ensure complete noising of images in order to prevent the use of methods of visual image processing.

One of the options to create the following modification is to combine elements of the RSA algorithm and binary operations in the software realization.

A primer numbers $P$ and $Q$ is called as elements of RSA algorithm, numbers $e$ and $d$ is from congruence $ed \equiv 1(mod\ \varphi(N)), N = P*Q$.

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## 2. Problem statement

Let suppose that the given image P has width $l$ and height $h$. It can be considered as a matrix of pixels [1].

$$< dtp_{ij} >_{1\le i\le n,\ 1\le j\le m}, \tag{1}$$

where $dtp_{ij}$ is the pixel with coordinates $i$ and $j$, $n$ and $m$ is number of points from width $l$ and height $h$. In general, $n$ and $m$ are dependent on $l$ and $h$, so such record is more correct:

$$n = n(l)\ \ im = m(h). \tag{2}$$

The matrix (1) corresponds to the matrix of pixel intensities [10]:

$$\mathbf{C}=\begin{pmatrix} c_{1,1} & \dots & c_{1,m} \\ \dots & \dots & \dots \\ c_{n,1} & \dots & c_{n,m} \end{pmatrix} \tag{3}$$

where $c_{ij}$ is the intensity value of grayscale images of the pixel $dtp_{ij}$. Therefore, there is such conformity [1]:

$$P = \mathbf{P}_{l,h} = \left[ pxl_{ij} \right]_{1\le i\le n(l),1\le j\le m(h)} \to \mathbf{C}=\left[ c_{ij} \right]_{1\le i\le n(l),1\le j\le m(h)} \tag{4}$$

1 byte is usually given to the gradation of brightness, moreover 0 is a black color, 255 is a white with maximum intensity.

Mathematically, contour in the image is the rupture of the spatial function of brightness levels in the plane of the image. Consequently, the separation of contour means the search of the sharpest changes, or maximums of the module of gradient vector [9]. This is one of the reasons why contours remain in image in the RSA system when they are being encrypted. Because the encryption here is based on the exponentiation module of some integers [12-14]. Herewith, on the contour and on the next pixels of the contour exponentiation value of brightness gives even bigger rupture [15-16].

## 3. Description of the developed algorithm using elements of RSA algorithm

### *3.1. Encrypting and decrypting by one row of image matrix*

Let $P$ and $Q$ be a pair of arbitrary prime numbers and $N = P*Q$. Encryption occurs elementally using the next transformation of image matrix $C$ elements:

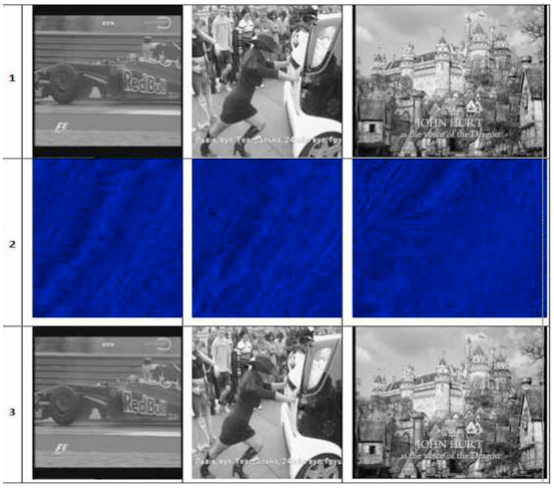



Fig. 1. a) Initial images; b) Encrypting images; c) Decrypting images

1. natural number $e < \varphi(N)$ is randomly chosen, and there is a natural $d$ that the congruent $ed \equiv 1 (mod\ (N))$ is executed;
2. the number $A = c_{ij} + Q + P + i + j - d$ is constructed;
3. the number $B \equiv A^e\ (mod\ N)$ is selected of the encrypted value of the $i$ pixel intensity, $i = 1,2,...,m$, $m$ is the number of line elements;

Decryption is carried in the opposite to the encrypting order after getting the number $B^d \equiv (A^e)^d (mod\ N)$ by the implementation of opposite operations to 3), 2), 1) items content. Results are shown in Fig. 1. Images for all experiments in the paper are taken from [17].

### *3.2. Encrypting and decrypting by one row of image matrix with extra noising*

Let $P$ and $Q$ be a pair of arbitrary prime numbers and $N = P*Q$. Encryption occurs elementally using the next transformation of image matrix $C$ elements:

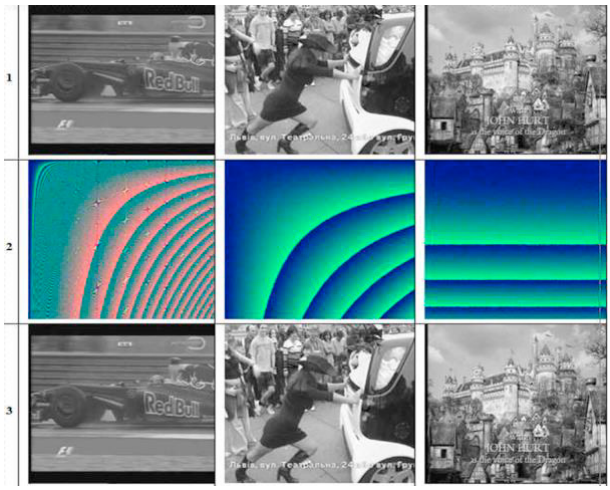



Fig. 2. a) Initial images; b) Encrypted images; c) Decrypted images

1. natural number e <$\varphi$ (N) is randomly chosen, and there is a natural $d$ that the congruent $ed \equiv 1(\text{mod } \varphi(N))$ is executed;
2. the number A = $c_{ij} + Q + P + i + j$ - d is constructed;
3. the number $C \equiv A^e \ (mod\ N) + f(i\ ,\ j)$ is selected of the encrypted value of the $i$ pixel intensity, $i= 1,2,...,m$, $m$ is the number of row's elements;

Decryption is carried in the opposite to the encrypting order after getting the number $(C\text{-}f(i,j))^{d} \equiv (A^{e})^{d}$ (mod $N$) by the implementation of opposite operations to 3), 2), 1) items content. Results are shown in Fig. 2. The following functions were chosen for encryption: $f(i,j) = I^{2}$, $f(i,j) = i*j$, $f(i,j) = j^{2}$ . The comparison of Fig. 1 and Fig. 2 shows that results of the encryption with extra noising differs from encryption without extra noising. There are no contours in both encrypted images. The initial and decoded images are only insignificant different in terms of brightness. The functions of extra noising $f(i,j)$ can be random integer functions and can also increase the cryptographic stability of these modifications additionally to the cryptographic stability generated by the RSA.

## 4. Description of the algorithm using bitwise operations

### *4.1. Algorithm for image encrypting using one row*

Let $P$, $Q$ – pair of arbitrary primes and $N = P * Q$, $\varphi(N) = (P - 1)(Q - 1)$. Encryption occurs element by element using the next transformation of the elements of the image matrix $\boldsymbol{C}$:

1. The natural number is randomly selected $e < \varphi(N)$ and finding such a natural $d$, so congruence performed $ed \equiv 1(\text{mod } \varphi(N))$.
2. If $i \equiv 0$ (mod 2), $1 \le i \le l$, then selecting randomly number $m \equiv (i + P)$ (mod 31) +1, and numbers are constructed $B \equiv m^{e}$ (mod $N$), $X = i*B*P$.
3. If $i \equiv 1$ (mod 2), $1 \le i \le l$, then selecting randomly number $m \equiv (i + Q)$ (mod 31)+1, and numbers are constructed $B \equiv m^{d}$ (mod $N$), $X = i*B*Q$.
4. Using a binary operation ^ bitwise exclude "OR" - constructing a number $a = c_{i,j} \wedge X$.
5. Each grade is highlighted $a_i$ number $a$ according to the following scheme:
   $a_1$= a & 01; $a_2$= a & 02; $a_3$ = a & 04; $a_4$= a & 010; $a_5$ = a & 020; $a_6$ = a & 040; $a_7$ = a & 0100; $a_8$ = a & 0200; $a_9$ = a & 0400; $a_{10}$ = a & 01000; $a_{11}$ = a & 02000; $a_{12}$ = a & 04000; $a_{13}$ = a & 010000; $a_{14}$ = a & 020000; $a_{15}$ = a & 040000; $a_{16}$ = a & 0100000; $a_{17}$ = a & 0200000; $a_{18}$ = a & 0400000; $a_{19}$ = a & 01000000; $a_{20}$ = a & 02000000; $a_{21}$ = a & 04000000; $a_{22}$ = a & 010000000; $a_{23}$ = a & 020000000; $a_{24}$ = a & 040000000; $a_{25}$ = a &

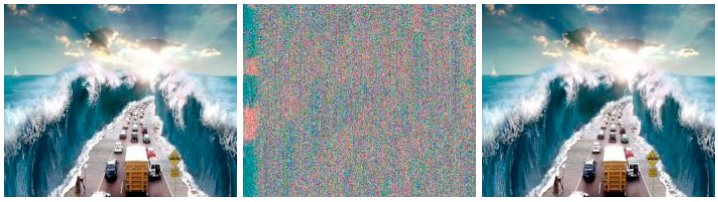
0100000000; $a_{26}$ = a & 0200000000; $a_{27}$ = a & 0400000000; $a_{28}$ = a & 01000000000; $a_{29}$ = a & 02000000000; $a_{30}$ = a & 04000000000; $a_{31}$ = a & 010000000000; $a_{32}$ = a & 020000000000,
where & - arithmetic "AND" operation.

6. A cyclic replacement is being performed $m + 1$ number's $a$ grades following the scheme: $k = a_{m+1}$ , $a_{m+1} = a_m$ , … , $a_2 = a_1$ , $a_1 = k$.
7. After performance step 5, image is encoded.
8. All the B numbers are written by following matrix: $V = \begin{pmatrix} b_{1,1} & \dots & b_{1,l} \\ \dots & \dots & \dots \\ b_{h,1} & \dots & b_{h,l} \end{pmatrix}$

*4.2. Algorithm for image dencrypting using one row*

Decrypting is performed with given numbers $e < \varphi(N)$ i $d$, $N = P*Q$, $\varphi(N) = (P - 1)(Q - 1)$.

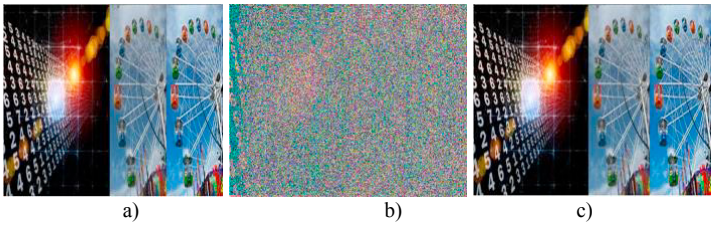
1. If $i \equiv 0 \pmod 2$, $1 \le i \le l$, then constructing a number $m \equiv B^d \pmod N$ and number $X = i*B*P$.
2. If $i \equiv 1 \pmod 2$, $1 \le i \le l$, then constructing a number $m \equiv B^e \pmod N$ and number $X = i*B*Q$.
3. Each $a_i$ number's digit is allocated by the following scheme:
$a_1$= a & 01; $a_2$ = a & 02; $a_3$ = a & 04; $a_4$ = a & 010; $a_5$ = a & 020; $a_6$ = a & 040; $a_7$ = a & 0100; $a_8$ = a & 0200; $a_9$ = a & 0400; $a_{10}$ = a & 01000; $a_{11}$ = a & 02000; $a_{12}$ = a & 04000; $a_{13}$ = a & 010000; $a_{14}$ = a & 020000; $a_{15}$ = a & 040000; $a_{16}$ = a & 0100000; $a_{17}$ = a & 0200000; $a_{18}$ = a & 0400000; $a_{19}$ = a & 01000000; $a_{20}$ = a & 02000000; $a_{21}$ = a & 04000000; $a_{22}$ = a & 010000000; $a_{23}$ = a & 020000000; $a_{24}$ = a & 040000000; $a_{25}$ = a & 0100000000; $a_{26}$ = a & 0200000000; $a_{27}$ = a & 0400000000; $a_{28}$ = a & 01000000000; $a_{29}$ = a & 02000000000; $a_{30}$ = a & 04000000000; $a_{31}$ = a & 010000000000; $a_{32}$ = a & 020000000000,
where & - arithmetic "I" operation.
4. A cyclic number's $a$ digits replacement is performed $m + 1$ by the following scheme: $k = a_{m+1}$ , $a_{m+1} = a_m$ , … , $a_2 = a_1$ , $a_1 = k$.
5. With ^ bitwise operation – exclusive «OR» for every digit - constructing a number $c_{i,j} = a \wedge X$.
6. After the 5th step image is decrypted.
Results with $P = 53$, $Q = 83$ are shown on Fig. 3.



a) b) c)

Fig. 3. Results of image encryption/decryption via the scheme by one row for P = 53, Q = 83: a) initial b) encrypted c) decrypted image

Results with P = 127, Q = 53 are shown on Fig. 4.



a) b) c)

Fig. 4. Results of image encryption/decryption via the scheme by one row for $P = 127$, $Q = 53$: a) initial b) encrypted c) decrypted image

The comparison Fig. 3 and Fig. 4 shows that the encryption with different values of primes $P$ and $Q$ does not differs significantly. There are no contours in both encrypted images. Initial and decrypted images only slightly vary in brightness. This approach can be used in different areas [18-22].

## 5. Conclusion

In this paper authors proposed the new encryption-decryption approach for grayscale and color images. Proposed algorithms, intended for encryption of grayscale images, are based on ideas, used in the basic RSA algorithm. The main difference is the additional use of bitwise binary operations. Based on the experimental evaluation it should be noted that: proposed modifications can be applied to any images, especially to images with well-defined contours; described modifications without reservations can be applied to color images; it is established that the influence of additional noise functions in the source image does not significantly affect the results of the method; the resulting image differs from the source one only by a small change in the brightness level; cryptographic strength of the developed modifications is provided as by RSA algorithm as well as by additional use of bitwise binary operations.

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